# The Izhikevich Neuron in Excitatory Cortical Neurons & Theta Neuron Implications

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A Mark Kramer inspired presentation.

# Original Paper

## Simple Model of Spiking Neurons

Eugene M. Izhikevich

***Abstract*—A model is presented that reproduces spiking and bursting behavior of known types of cortical neurons. The model combines the biologically plausibility of Hodgkin–Huxley-type dynamics and the computational efficiency of integrate-and-fire neurons. Using this model, one can simulate tens of thousands of spiking cortical neurons in real time (1 ms resolution) using a desktop PC.**

***Index Terms*—Bursting, cortex, Hodgkin–Huxley, PCNN, quadratic integrate-and-fire, spiking, thalamus.**

https://www.izhikevich.org/cv/index.htm

Izhikevich, E. M. Simple model of spiking neurons. IEEE Trans Neural Netw 14, 1569–1572 (2003).

# Motivation

Create a biophysical model that captures a wide variety of cortical activity – like bursting – while remaining relatively mathematically simple.

How can we create an efficient model that accurately mimics neural activity biologically (HH) while remaining computationally efficient (LIF)?

… Izhikevich Neuron!

But what is it?

# Model System

How does **voltage** change in time?

$$v' = 0.04v^2 + 5v + 140 - u + I$$

$$u' = a(bv - u)$$

How does the **membrane recovery variable** change in time?

$$\text{if } v \geq 30\,\text{mV}, \text{ then } \begin{cases} v \leftarrow c \\ u \leftarrow u + d. \end{cases}$$

Model for a single neuron

Representation of firing patterns changing

What is a, b, c, and d?

# Model System

Input current

$$v' = 0.04v^2 + 5v + 140 - u + I$$

$$u' = a(bv - u)$$

Sensitivity of recovery variable to voltage

Time scale of recovery variable

After-spike reset voltage (where v goes after spike) – K+ conductance dependent

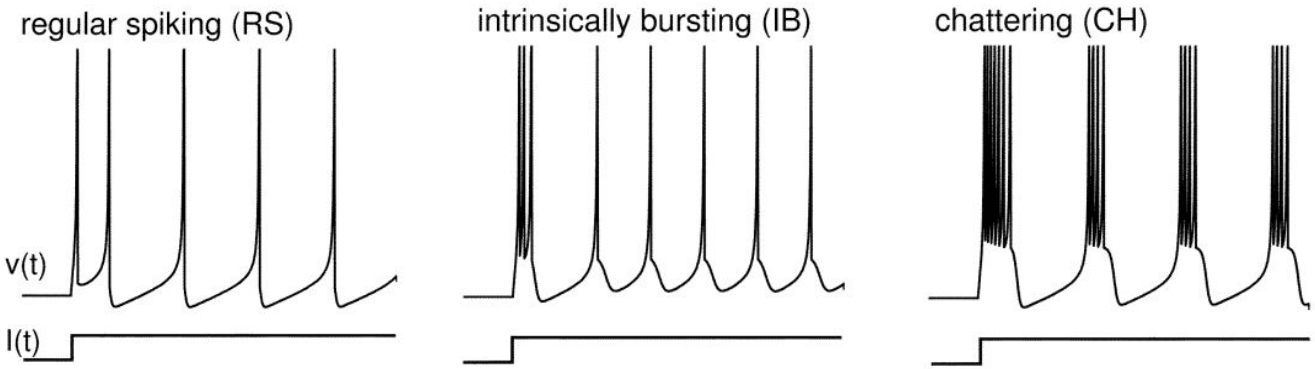
$$\text{if } v \geq 30\,\text{mV}, \text{ then } \begin{cases} v \leftarrow c \\ u \leftarrow u + d. \end{cases}$$

After-spike jump in recovery (how much u increases per spike) – Na+ and K+ conductance dependent

# What now?

Model spiking and bursting activity in excitatory cortical neurons!

Excitatory cortical neurons mainly exhibit 3 types of behavior…




Modulate parameters to model this in Izhikevich Neuron

```python
# Simulate the model
def izhikevich_sim(a, b, c, d, I_fn, T=300.0, dt=0.1, v0=-65.0): # All inputs taken in the function

    # Create arrays to store changing variables over time
    n = int(T / dt) # How many time steps
    t = np.arange(n) * dt
    v = np.zeros(n)
    u = np.zeros(n)
    I = np.zeros(n)

    v[0] = v0  # Initialize voltage by calling function input
    u[0] = b * v0  # Intialize membrane recovery variable

    for i in range(n - 1):  # Go through each time step and find change in voltage and membrane recovery variable (dv and du)
        I[i] = I_fn(t[i]) # Input current is current function at time i

        dv = 0.04 * v[i]**2 + 5 * v[i] + 140 - u[i] + I[i]  # Directly from model, change in voltage over time
        du = a * (b * v[i] - u[i])  # Also directly from model, change in membrane recovery variable over time

        # Uses above calculations to calculate next step
        v[i + 1] = v[i] + dt * dv
        u[i + 1] = u[i] + dt * du

        if v[i + 1] >= 30:  # Reset voltage and update variables according to parameters c and d
            v[i] = 30  # show spike peak in plot
            v[i + 1] = c
            u[i + 1] += d

    I[-1] = I_fn(t[-1]) # Input current now at very end of sequence
    return t, v, u, I # Return variables that change in time
```

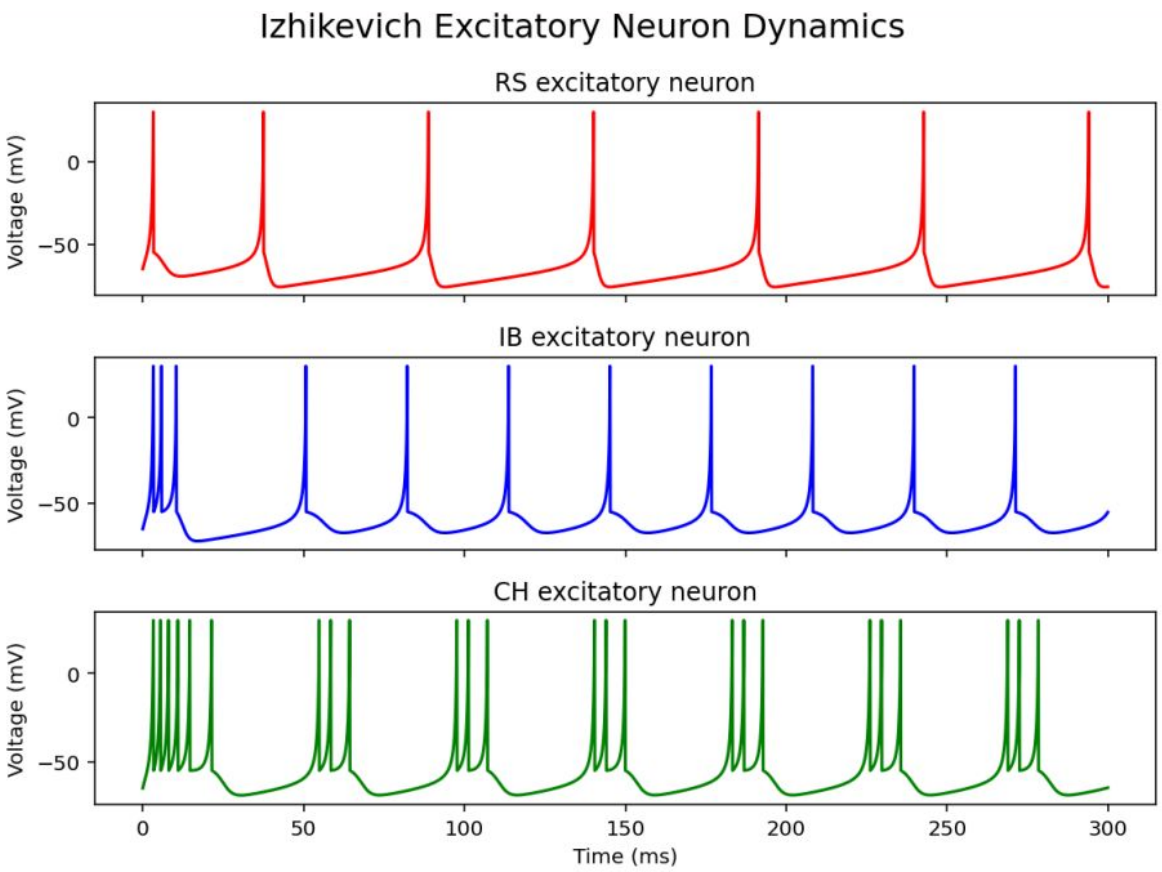
Lines that plot behavior are not shown (below).

# Code Results

Values of a, b, and c all the same for each behavior, but not d!

a = 0.02, b = 0.2, c = -55

d = 10
d = 4
d = 1.75



```python
# How long the function runs for
def step_current(start=50, stop=250, amp=10):
    return lambda t: amp if start <= t <= stop else 0.0

# Three different spiking patterns for excitatory neurons
if __name__ == "__main__":
    # Regular-spiking excitatory interneuron (RS)
    rs_params = dict(a=0.02, b=0.2, c=-55, d=10)

    # Intrinsically bursting excitatory interneuron (IB)
    ib_params = dict(a=0.02, b=0.2, c=-55, d=4)

    # Chattering excitatory interneuron (CH)
    ch_params = dict(a=0.02, b=0.2, c=-55, d=1.75)

    I_fn = lambda t: 10  # Setting current to constant

    # Call the ihikevich model and input parameters we set for each type of behavior
    t_rs, v_rs, u_rs, I_rs = izhikevich_sim(**rs_params, I_fn=I_fn)
    t_ib, v_ib, u_ib, I_ib = izhikevich_sim(**ib_params, I_fn=I_fn)
    t_ch, v_ch, u_ch, I_ch = izhikevich_sim(**ch_params, I_fn=I_fn)
```

Code for different firing activity.

# Parameter d

As we decrease the value of d, more periodic bursting…

d = 10
d = 4
d = 1.75

But why? What is d?

After-spike reset value of u caused by slow high-threshold Na+ and K+ conductances

$$\text{if } v \geq 30\,\text{mV}, \text{ then } \begin{cases} v \leftarrow c \\ u \leftarrow u + d. \end{cases}$$

# Parameter d

$$\text{if } v \geq 30\,\text{mV}, \text{ then } \begin{cases} v \leftarrow c \\ u \leftarrow u + d. \end{cases}$$

Hit peak of action potential, recovery variable u is changed by d

u is membrane recovery variable, changes with time

d large value (10) → large u → more time for neuron to recover
d small value (1.75) → small u → less time for neuron to recover

# Biological sense

Let's put it all together in the context of excitatory cells:

d = Spike-triggered increase in recovery variable (u) → how does the neuron "push back" after each spike?

- Slow k+ currents (e.g. adaptation)
- Mechanisms responsible for spike-frequency adaptation

After each spike:

- u increases by +d
  - Neuron temporarily becomes less excitable
  - Period for refraction
- Lower d → weaker adaptation → more bursting
- Larger d → stronger adaptation → regular spiking

# Theta Neuron

Model that collapses dynamics into single variable theta ($\theta$)

Spiking = $\theta$ wraps around the circle ($-\pi \rightarrow +\pi$)

Behavior is dependent on input variable “a”

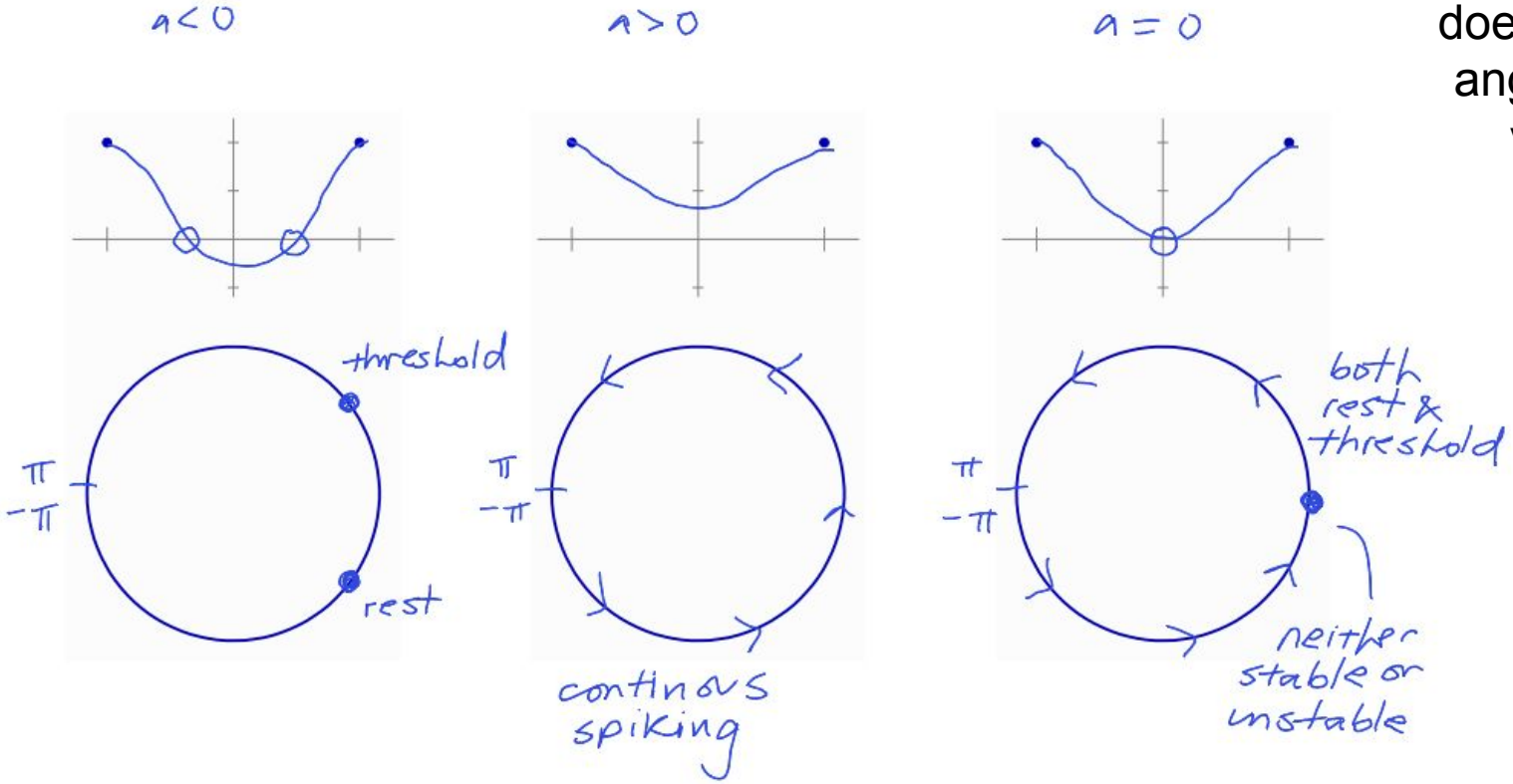
- $a < 0 \rightarrow$ excitable
- $a > 0 \rightarrow$ continuous firing

Dynamics controlled by how fast $\theta$ moves around the circle

Parameter a shifts system toward or away from continuous firing

$$\frac{d\theta}{dt} = f(\theta) = (1 - \cos\theta) + (1 + \cos\theta)\, a$$

But Izhikevich neuron doesn't have angle nor a value!



# Theta Neuron With Izhikevich

Izhikevich doesn't have a natural unit-circle trajectory…

but we can construct one!

Because of reset in model, we see a sharp jump in phase

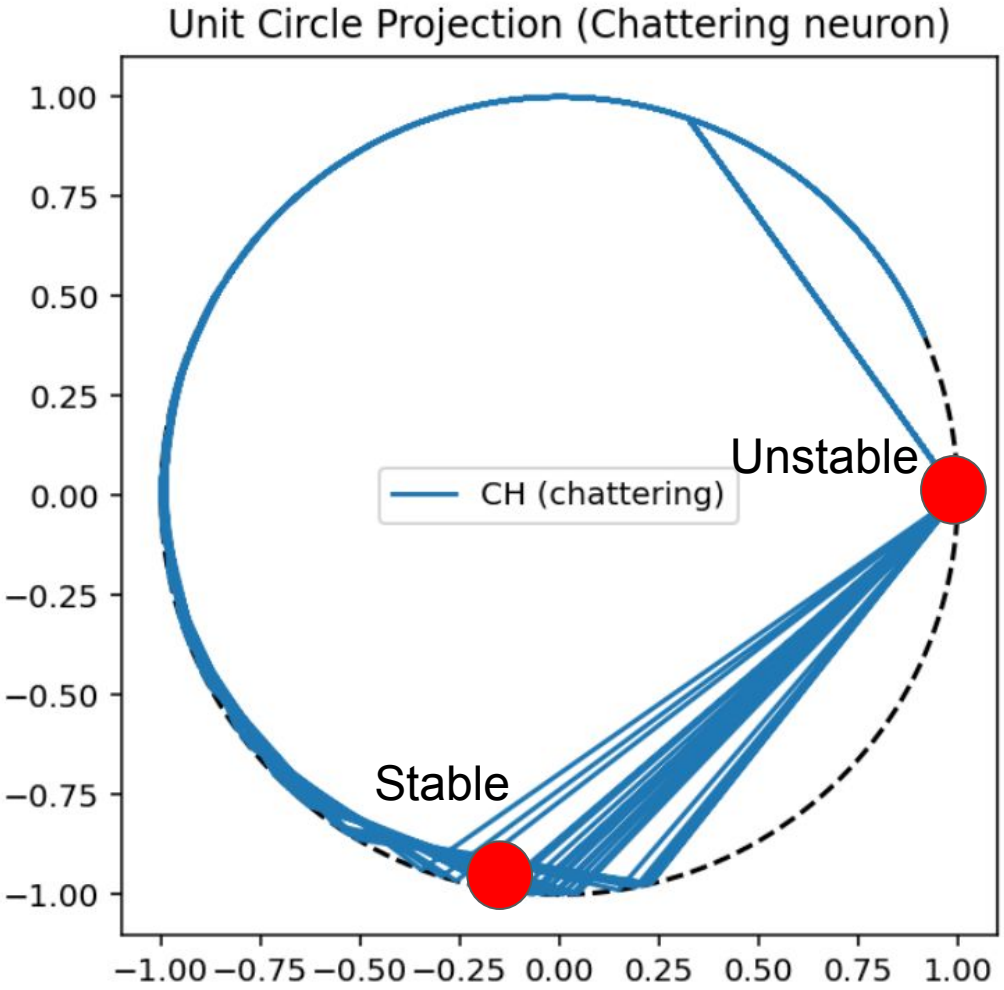
→ not a perfectly smooth rotation

$$\text{if } v \geq 30\,\text{mV}, \text{ then } \begin{cases} v \leftarrow c \\ u \leftarrow u + d. \end{cases}$$

What does this look like in a chattering neuron –
the most extreme form of bursting?

# Theta Neuron With Izhikevich

| CONCEPT | THETA NEURON | IZHIKEVICH |
|---|---|---|
| Fast Variable | θ (phase) | V (voltage) |
| Input Drive | a | I |
| Spike Reset | Wrap around circle | V reset + u jump |
| Adaptation | N/A | u |
| Stopping Firing | Lower a | Higher u (effecting d) |




Goes from continuous firing to jumping = spike reset in voltage

Dense area of kinks in rotation is the frequent resets in bursting

Stable and unstable states?

Stable → many jumps to this area
Unstable → jumps away from this area

```python
# Create theta model for chattering neuron
# Get min and max voltage values for CH
v_min, v_max = np.min(v_ch), np.max(v_ch)

# Create units along pi respective to min and max values
theta_ch = 2*np.pi * (v_ch - v_min) / (v_max - v_min)
x_ch, y_ch = np.cos(theta_ch), np.sin(theta_ch)

# Create unit circle
plt.figure(figsize=(5,5))
circle = np.linspace(0, 2*np.pi, 200)
plt.plot(np.cos(circle), np.sin(circle), 'k--')

# Plot voltage trajectory of chattering neuron on circle
plt.plot(x_ch, y_ch, label='CH (chattering)')
plt.gca().set_aspect('equal')
plt.legend()
plt.title("Unit Circle Projection (Chattering neuron)")
plt.show()
```

# Caveats

Cannot truly represent a two-dimensional system on a unit-circle

In reality, to represent bursting, the theta model needs an adaptive system added on that causes activate to oscillate back and forward

→ but can’t show that in a unit circle visually, just have to pretend it’s there working in the background

→ would need a 3-D system showing voltage trajectory as well as how adaptive system works alongside of it

# Summary

- Izhikevich neuron is good at combining LIF and HH to producing bursting and spiking activity
- Parameter d seems to play an important role in frequency of bursting
- Theta neuron can be used to plot Izhikevich voltage trajectory, but has some caveats

# Thanks!
# Any questions?